

INTERNATIONAL TRAINING ON MULTI-AGENTS SYSTEMS FOR NATURAL RESOURCES MANAGEMENT (2 weeks)

About Multi Agents Systems

Recent advances in the field of Distributed Artificial Intelligence (DAI), and the advent of more powerful microcomputers, are raising the level of interest in multi-agents systems for various applications. This is particularly true in Natural Resources Management (NRM) since multi-agents systems constitute powerful tools for studying interactions between societies and their environment. They can also provide solutions for scaling issues.

Development of the multi-agents systems approach is closely related to the problem of complexity (multiple scales and organization levels, multiple agents and viewpoints, etc.) and the related search for simple representations of the real world through modeling. Identifying the conditions that allow the co-viability of environmental/resources dynamics on one hand and socioeconomic dynamics on the other hand is increasingly seen as a key issue for sustainable NRM because it focuses on the interactions between various agents (individual or collective, each one having a certain autonomy) that are acting in a given environment. The multi-agents systems approach seems well adapted to dealing with this core problem.

A multi-agents system is made up of a set of agents evolving in a common environment. These agents communicate and interact. The system is characterized by its structures as well as its control

and communication methods. These artificial multi-agents systems are the subject of research in informatics. They require the use of computers or networks of computers to develop metaphors (social, biological, physical) expressing a vision of the real world as sets of interacting entities. Finally, they provide a way to simulate complex real situations, phenomena, or organizations.

Multi-agents systems are particularly efficient in exploring the consequences of interactions between components of a complex system:

- to implement a bottom-up approach for understanding realities,
- to study the dynamic linkage between the behavior of a whole entity and the behavior of its parts,
- to elucidate emerging phenomena,
- to study the viability of a system based on interactions between its parts.

Different types of applications for the simulation of complex systems can be mentioned:

- simulation of aggregate or reproductive behavior,
- simulation of the emergence of a specialization,
- simulation of social phenomena, of interactions between different kinds of users exploiting a common resource, of urban growth,
- simulation of runoff, etc.

Objectives

- To introduce multi-agents systems (MAS) and review the state of the art in applying MAS to several key scientific disciplines, with an emphasis on NRM issues;
- To enable participants to develop their own simple MAS application by constructing and operating a MAS on a topic of their choice;
- To identify future opportunities for developing the use and application of the MAS approach to key NRM issues in the region.

Course Duration

The duration of the course will be for two weeks.

Course Outline

- Introduction to the study of dynamics systems
- Preliminary exercises on Smalltalk computer language
- Introduction to the game theory
- Introduction to cellular automata
- Second range of exercises on Smalltalk language
- Introduction to multi-agents systems (MAS)
- Theoretical approach to MAS, Part 1
- Presentation of the common-pool resource and multi-agents systems (CORMAS) platform
- Theoretical approach to MAS, Part 2
- Practical exercise on using CORMAS platform for MAS
- Theoretical approach to MAS, Part 3
- State of the art in MAS (Part 1): Ecology and geography
- More practical exercises on CORMAS platform for MAS
- State of the art in MAS (Part 2): Game theory
- State of the art in MAS (Part 3): Economics

- Construction of a prototype MAS
- State of the art in MAS (Part 4): Agronomy
- Practicing on MAS: Irrigated systems
- The use of models
- The “Shadoc” role playing game (irrigated systems)
- Preparation of the prototype presentation
- Presentation and discussion of the prototypes built by participants
- Evaluation
- Plan for future training and collaboration on MAS

Course Approach and Structure

The course will be using a combination of lectures, practical exercises, presentations, and discussion.

Participants

The class will be limited to a maximum of 16 trainees. Participants should be professionals or graduate students working on complex issues in the field of integrated natural resources management at either conceptualization or implementation stages. They should be familiar with the PC environment, but no special skills in mathematics or computer programming are required, even if the training course is heavily computer-based

For more information:

Dr. François Bousquet, Dr. Christophe Le Page

ERE-TERA, CIRAD

TA60/15 - 73, av. Jean François Breton

34938 Montpellier Cedex 5 – France

bousquet@cirad.fr, lepage@cirad.fr

<http://www.cirad.fr/presentation/programmes/espace/cormas/eng/index.shtml>